Machine Learning Embedded in Distribution Network Relays to Classify and Locate Faults

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***Abstract*—Distribution system relays equipped with machine learning (ML) algorithms can improve protection operations by performing an automated and self-learning monitoring and decision making analysis. This work evaluates the potential for a Support Vector Machine (SVM) classifier to be embedded inside each relay and provide decision making support based on the analysis of typical electrical performance measurements. The experiment, described here, used simulated data from the IEEE 123 bus feeder to understand the SVM algorithms abilities. The results showed that the algorithm deployed inside each relay could accurately classify three fault conditions that occur anywhere on the feeder, estimate the fault's region, and define a specific action for the relay switch. This initial assessment indicates that advanced, data-driven relay analysis could provide value in a typical feeder.**

***Index Terms*—relays, machine learning, faults classification**

I. Introduction

Distribution network relays programmed to isolate faults using machine learning (ML) algorithms provide advanced awareness of system conditions. During tumultuous times when the grid is experiencing a fault the ML provides a quick and independent analysis to effectively isolate the issue. Existing relays use differential, distance, and other approaches to segregate faults and maintain system integrity [1]. These devices scattered throughout the network often work together to understand the fault and extent of the isolation required. The collaborative effort between the relays involves an iterative process of switching, where upstream relays open and close multiple times to see if a downstream relay isolates the issue. Instead of relying on these type of interactions, ML could improve the individual capabilities and allow each device to confidently know the location and react instantly.

ML algorithms embedded inside each relay at the grid edge can improve system safety by providing a quick and accurate response. Additionally, ML solutions are necessary as DER, such as solar photovoltaics (PV), become more prevalent or a fast adoption of new Electric Vehicle (EV) loads occurs; distribution systems may also undergo changes to the network topology to provide short and long-term resilient operations, which requires relays to make necessary adjustments to accommodate. ML inside each relay supports these efforts by providing a more advanced system for detecting and reacting to fault conditions.

The advanced real-time analytics provided by the ML on each relay, allows for quick decision making and identification of the fault region. Unlike existing approaches, which rely on protection engineers to set the rule-based thresholds that determine the triggering of the relay, a ML algorithm can understand normal and fault conditions through a data driven analysis. The advanced learning allows for the quick determination of the fault to avoid slow or inaccurate determination of the area to segregate.

Recent literature examined how ML algorithms can improve distribution system protection approaches. Past exploration used ML for detection and diagnosis of symmetrical and power swing faults in transmission systems [2]; other transmission system work focused on how Support Vector Machine (SVM) algorithms can correct the maloperation of relays and potentially compensate for missing data [3] and compared the accuracies associated with different features [4]. The present work instead focused on distribution system protection relays using the SVM algorithm.

Distribution network studies considered the implementation of different ML approaches including a Reinforcement Learning (RL) algorithm which improved the failure rate of relays simulated in the IEEE 123 bus feeder [5]. A study, using a GridlabD IEEE 123 bus feeder model, compared an SVM-based relay with an over-current approach under two fault impedance scenarios using active and reactive power data and required communications between relays and/or a centralized computer [6]. In contrast, the present work evaluated three potential fault types that occurred under a range of impedance values, and the ML embedded in each relay collected and analyzed data without help from its peers or a central computer.

Past work used various levels of data frequency. For instance, a microgrid protection research endeavor evaluated the potential for wavelet transform and SVM analysis to improve protection results [7]. Similar work used high frequency data and coordinated control to define relay states in a distribution system [8]. This work used lower frequency data at about 1000 Hz. Others also leverage low frequency data to diagnose faults with an SVM algorithm [9] and identify their locations [10], [11] for centralized oversight. Yet, this work evaluates the potential accuracies of ML driven relay decisions that does not require knowledge of its neighbor's sensor readings or help from a centralized server.

To explore the potential for an independent and adaptive fault analysis approach, this paper highlights the ability of a Support Vector Machine (SVM) to assess individual relay data within the IEEE 123 bus feeder. Unlike Lin et al, that uses a Support Vector Data Descriptor to estimate distances [12], this work uses the electrical performance data measured within each relay to determine fault types and locations. The simulation experiment emulated a feeder using Matlab Simulink; the feeder included ten relays distributed throughout the system; the model generated current and voltage data over a one-year period and included three types of faults at 23 different locations. Data captured by each of the relays during normal and fault conditions provided inputs into the SVM, which classified fault types and locations.

This paper adds to previous work that used an SVM or other ML algorithms for relay protection decisions by providing a new analysis perspective focused on the following items:

1) All relay's operate without communication with peers or a central command center to make decisions.
2) A settingless (or binary), reliable relay decision that depends on the SVM ability to identify whether the fault resides within the relays region of interest or not.
3) Accurate identification of the faulted region by individual relays.

The analysis used data generated by a simulation of the IEEE 123 bus feeder subjected to various fault conditions at different locations.

II. Simulation Data

A. Distribution System

The simulation effort used the IEEE 123 bus feeder, originally created as a demonstration model for comparison testing [15], to emulate the normal and fault conditions. Recently researchers used the demonstration circuit to investigate voltage-load sensitivity [16], a feeder reconfiguration study [17], renewable energy hosting capacity [18], and many others. The feeder, depicted in Fig. 1, included a three-phase backbone rated at 4.16 kV that supported a range of loads. During the simulation the model disabled the regulators so that tap changes did not alter the system voltage.

The IEEE 123 bus feeder used in this work included ten relays that resulted in the creation of eight regions as shown in Fig. 1. The location of the relays was based on the methodology defined by Reimer et *al.* [19]. The region directly connected to a relay on the opposite end from the substation was defined as its region of interest.

B. Fault Conditions

This work considered three fault types: (1) Single Line to Ground (SLG), (2) Line to Line (LL), and three phases to ground (3LG). The three fault types provided interesting results because the feeder had an unbalanced configuration. The simulation effort implemented these faults at different times, and included various levels of impedance.

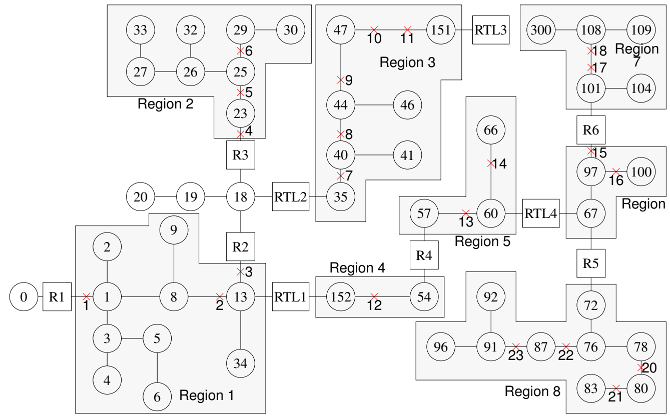


Fig. 1. Modified IEEE 123 bus distribution system includes 91 loads protected with a total of 10 relays. Relay locations are indicated with the letter R and RTL (relay tie lines). The simulation included three fault types at the 23 different locations inside 8 regions. The node labeled with a zero represented the systems substation.

C. Simulation Process

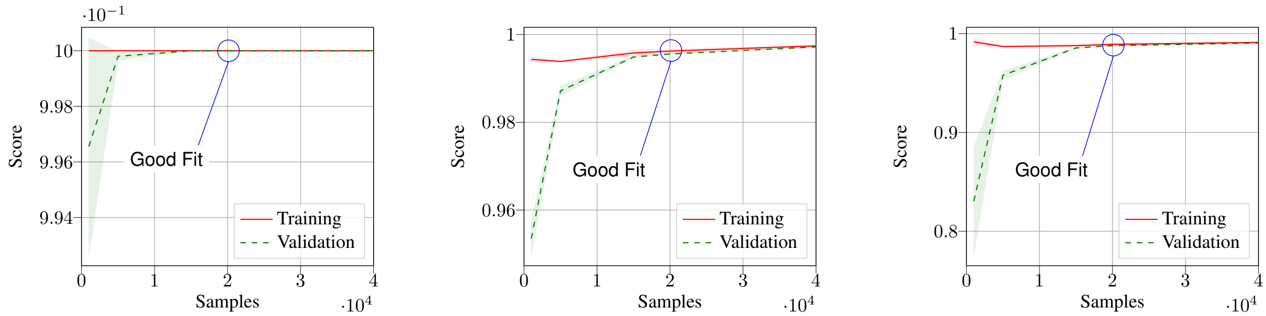
A year long simulation of the IEEE 123 bus feeder in Simulink emulated normal and multiple fault conditions (described in Sec. II-B). The simulation allowed for faults to be applied at one of the 23 locations indicated by the red crosses in Fig 1. The emulation of the fault conditions included a variation in the fault impedance (i.e. random value between 0.1 to 5 ohms) and the fault duration, which varied between 0.4 and 3 seconds. High impedance faults (HIF) on distribution feeders include currents magnitudes anywhere from zero to less than 100 amperes, these fault currents are below traditional over-current relay pickups [20] and not considered in this work.

III. Support Vector Machine Algorithm

The SVM algorithm, created by Cortes and Vapnik [13], learned the different classes by separating the data with a multi-dimensional hyperplane. The hyperplane was created by maximizing the minimum distance to the training points closest to the plane. This was accomplished by mapping the input vectors into a high dimension feature space; in this space, a non-linear surface separated the data into the respective classes. The SVM algorithm required two free parameters to be preset: the regularization parameter (C) defined the degree of importance associated with avoiding misclassification and gamma (γ) acted as the kernal's coefficient. In this case a radial basis kernal was used to separate the data [14]. This work set the SVM to have a radial basis kernel, a C parameter equal to 10,000, and a γ of 100.

IV. Fault Analysis

The fault analysis focused on the potential for an SVM classifier embedded inside a relay to detect, react, and locate system faults. This particular approach considered each data point individually and not the time-series dependancies. The assessment used simulation data from the model describe in Sec. II to evaluate normal and fault conditions; the simulation



(a) Fault classification

(b) Open/Close decisions

(c) Fault region classification

Fig. 2. Validation of the training samples involved training and testing on various levels of data samples for each of the three applications. The analysis found that for fault classification, open/close decision making, and fault region classification a sample quantity of 20,000 avoided under and over model fitting.

generated 3.6 million data points that included 55.7% normal, 29.6% SLG, 10.2% LL, and 4.5% 2LG. The analysis however, did not consider all of the data, and instead select a random subset based on an assessment of the learning curves which considered the tradeoff between an under- and over-fit model for each analysis type (i.e fault classification, breaker decision, and fault region determination).

The first step in the analysis involved the classification of the fault type. Results from this analysis were used to inform the next step, which determined if the fault occurred inside the relay's region. This emulated the situation where the SVM acted as a decision maker and dictated the opening and closing of the breaker. The final analysis used the local data to define the particular region where the fault occurred.

A. Data Input Types

The SVM algorithm used data measured and archived inside typical modern relays. The measured current and voltage on the three phase electrical system can be reduced mathematically to symmetrical components described as positive, negative, and zero sequence voltage (V_1, V_2, and V_0) and current (I_1, I_2, and I_0) [21]. Each of the six components were provided as inputs into the SVM to classify the fault types, understand the faults location, and determine if the fault resided within the relays protection region of interest or not.

B. Evaluation Metrics

The confusion matrix (CM) defines the performance of the learning algorithm by comparing the predicted and actual outcomes in a matrix format. The format allows for the review of the True Positive (TP) , False Positive (FP), True Negative (TN), and False Negative (FN) [22]. This provides a thorough review of the algorithm's abilities. In contrast, the proportion of correctly classified (or accuracy) does not highlight the overall algorithm capabilities in significant detail. The matrix results, described in Secs. V-A and V-C, normalize the predicted and actual outcomes by dividing each value (TP, TN, FN, FP) by the total number of predictions (or sum of the column).

To explain the algorithms ability to make switching decisions, the CM elements were described in a modified format that separated the True Positive Rate (TPR), False Positive Rate (FPR), True Negative Rate (TNR), and False Negative Rate (FNR) into different scenarios (Sec. V-B). The modified CM described the overall results, the algorithms ability to understand the location of faults in its own zone, in a different zone, and when no faults were present. Faults defined to be "In Zone" meant that it occurred on the opposite side of the relay from the substation. For example, an "In Zone" fault for relay RTL4 could occur in Regions 6, 7, and 8. "Out of Zone" faults were located on the substation side of the relay, and include Regions 1, 2, 3, 4, and 5 for relay RTL4.

V. Results

The three applications for the SVM algorithm embedded inside each relay all used the same input features (V_1, V_2, V_0, I_1, I_2, and I_0), but had different target variables. The target values depended on the particular application. The fault classification estimates used the fault type variable; identification of the fault's location required the faults region as an input; relay decision making was a two-class problem, where the region directly in-front of the relay was labeled as a 1 and any other region was 0.

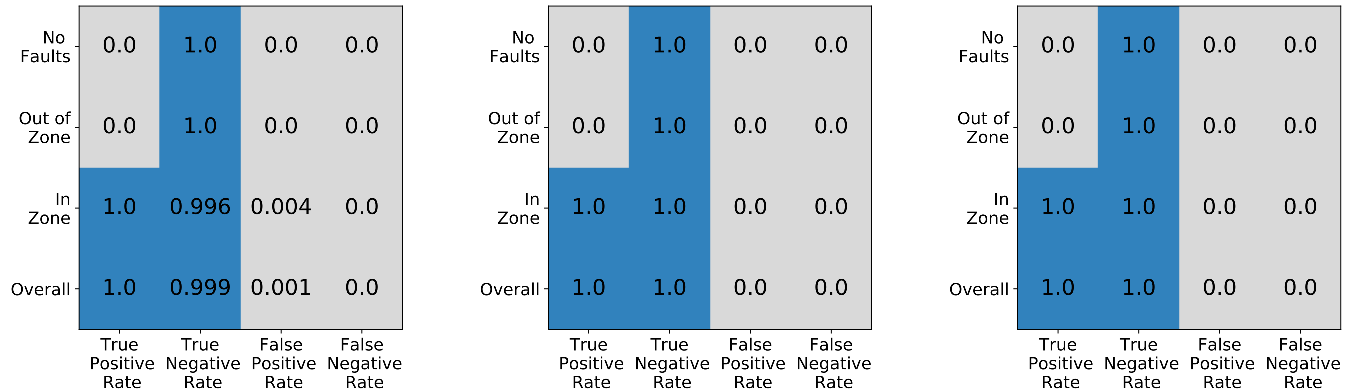
To define an appropriate training size, the analysis included a validation process. The process, which iterated through different samples sizes and computed training and validation scores, found a sampling size that avoided under- and over-fitting of the model. The learning curve results, shown in Fig. 2, describe when a low number of training samples result in high variance. High variance occurs when the validation and training scores have a significant difference. A good model fit occurred when the training and validation scores converged close to 20,000 samples for each of the three analysis applications.

A. Fault Classification

The classification of faults proved to be a simple endeavor for all ten relays. The SVM algorithm that assessed the sequence currents and voltages measured by all of the relays could classify each of the 23 faults with 100% accuracy. Furthermore, the TPR and FPRs were computed to be 100% and 0% respectively.

B. Relay Switch Decisions using Machine Learning

The experiment found that ML could potentially make switching decisions inside of protection relays. This binary



(a) Relay decision results for Fault Type SLG (b) Relay decision results for Fault Type LL (c) Relay decision results for Fault Type 3LG

Fig. 3. Example results describing the accuracy of the SVM inside the RTL4 relay. The matrix describes the results for data without faults, data of faults outside and inside its zone, and the overall performance. When assessing the type LL and 3LG faults, the SVM algorithm performed perfectly with a TPR and FPR of 100% and 0% respectively. During a SLG fault event, the SVM algorithm was not perfect but had a high TPR of 99.9% and a low FPR of 0.1% overall.

decision, which defined whether the switch should be opened or remain closed, depended on the SVM's ability to understand where the fault occurred. Each relay protected the network by monitoring a single region as described in Fig. 1. For example, this setup intended for relay R4 to open only when a fault occurred in Region 5 or relay RTL4 to respond to faults in Region 6 only and not react when faults happen in Regions 7 or 8.

The decisions made by the SVM algorithm in each relay depended on training data produced for each fault type. Since the fault classification analysis produced perfect results, the analytics inside the relay could first define the type and then identified its location. Providing specific training for each fault condition decreased the complexity of the problem and improved the overall accuracy.

In this analysis, the SVM algorithm avoided the misclassification of normal operating points (i.e. No Faults) for each of the relays when subjected to all fault conditions. Example results, shown in Fig. 3 for relay RTL4, describe the TNR and FNR to be 100% and 0% respectively for each of the fault types.

The SVM algorithm handled the "Out of Zone" and "In Zone" fault events (explained in Sec. IV) with a high degree of accuracy. "Out of Zone" fault events, which did not require the relay to respond, had a TNR of 100% for each of the fault types. Faults events defined as "In Zone" also had a high degree of accuracy and the SVM produced a TPR of 100%. However, the SVM analysis did not result in an ideal TNR or FPR for events that included the SLG fault type. In this case, the FPR was still very low and TNR was high at 99.6% and 0.4% respectively. The "In Zone" results were found to not include any errors for fault types LL and 3LG.

Overall the SVM algorithm performed very well at making switching decisions. Some errors did occur when subjected to SLG fault types, but the FPR was still very low at 0.1%. In the other fault cases, the SVM generated FPR of 0% overall. These results provide some evidence that actual protection relays could operate using ML algorithms. The ML algorithms could also provide other services or decision such as the quick determination of reconfiguration strategies. This would require the relays to autonomously and quickly understand fault locations.

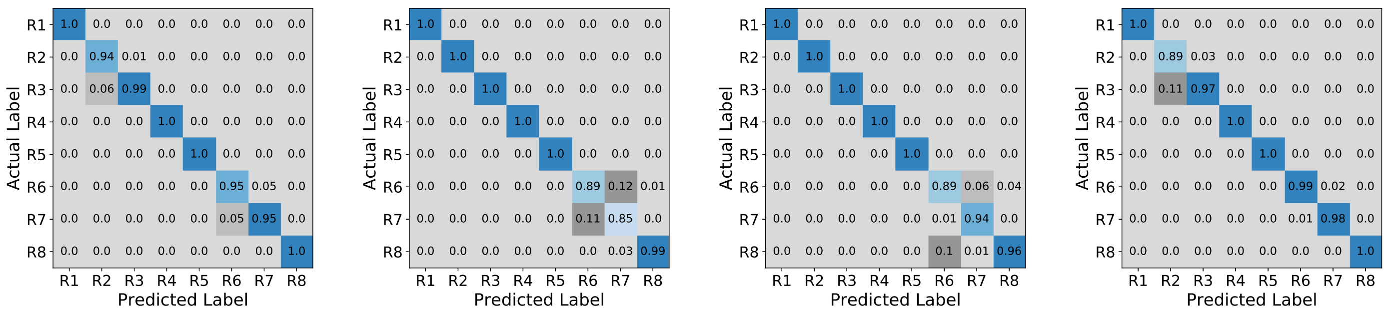
C. *Fault Location Identification*

This work also assessed the ability of the SVM to accurately identify fault locations from any relay on the network. The eight regions, shown on Fig. 1, represented the electrical lines in-between the relays. The results shown here describe the abilities of the tie line relays (RTL1, RTL2, RTL3, and RTL4) to identify the region where the fault occurred. These relays provide a mechanism for the reconfiguration of the network. And, if each of them could accurately estimate the fault locations they could potentially perform autonomous reconfiguration control actions. This paper does not evaluate or speculate on the methodology for reconfiguration, but instead focuses on the SVM's ability to estimate the fault location which could be a critical input for the control system.

Overall the SVM algorithm embedded inside the relays were successful, but had difficulties understanding the difference between some areas. Fig. 4 describes the results produced by the algorithm embedded inside each of the tie line relays. The analysis performed within the RTL1, RTL2, and RTL3 (Figs. 4(a), 4(b), and 4(c)) had a high TPR and low FPR for identifying faults in Regions 1, 3, 4, 5, and 8. The three relays had trouble distinguishing between faults in Regions 6, 7 and 8. RTL1 and RTL4 (Fig. 4(d)) had some trouble with understanding the difference between Regions 2 and 3. The algorithm had a higher FPR trying to distinguish the difference between regions close to one another especially at a split in the network topology.

VI. Conclusion

The experiment found that an SVM embedded inside each relay could accurately classify faults, determine tie line switch positions and estimate fault locations. The results show that selectivity and sensitivity of the relay are improved with an SVM embedded in the relay. The relays were able to successful operate in an autonomous manner and not require



(a) RTL1 (b) RTL2 (c) RTL3 (d) RTL4

Fig. 4. Confusion matrices describing classification of fault regions (labeled w/ R) results for the tie line relays. Results show successful classification of fault regions. Some regions were harder than others for some of the feeders. For example, RTL1 and RTL2 had trouble distinguishing between Regions 6 and 7.

its peers or a centralized command center for help. Future work in this area will study its ability to operate under more realistic conditions using a power hardware-in-the-loop simulation environment. This work will also provide a basis for future reconfiguration work where the individual relays can understand fault locations and provide inputs into specific control schemes. The hope is that it can eventually be used to protect systems with DERs and active reconfiguration needs and require little to no intervention from humans.

Acknowledgment

This material is based upon work supported by the U.S. Office of Electricity. The paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

Sandia National Laboratories is a multimission laboratory managed and operated by National and Engineering Solutions of Sandia, LLC., a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.

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